WasteSort Monday Speaking Guide

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Suggested length: about 4 minutes

Read this once, then explain the ideas in your own words. The slide notes contain the same script.

1. WasteSort and basic machine learning

Good morning. I'm Lordy Allen Reyes. Today I'll explain the basic ideas behind our WasteSort project: artificial intelligence, machine learning, the main learning types, and how we use supervised image classification. Our task is to predict the material of one waste item from a photograph. This presentation focuses on the concepts behind that task.

2. Artificial intelligence and machine learning

Artificial intelligence, or AI, is the broad field of building systems that perform tasks such as recognizing images and making decisions. Machine learning is one approach within AI. Instead of writing a separate rule for every possible image, we train a model using examples. A model is the learned mathematical system that makes a prediction. Deep learning is a type of machine learning that uses multilayer neural networks.

3. Common learning types

In supervised learning, examples include known answers called labels. For WasteSort, a photo can have the label plastic. In unsupervised learning, the system looks for patterns without target labels, such as grouping similar customers. In reinforcement learning, an agent learns from rewards after actions, such as a game-playing system improving its strategy. WasteSort uses supervised learning because our training images already have material labels.

4. Practical machine learning applications

Machine learning appears in several everyday tasks. A spam filter classifies an email into a category. A travel-time model predicts a number, so that task is regression. A clustering model groups similar examples without predefined target labels. WasteSort belongs to classification because its answer is a material category. Writing a category as a number in code does not turn the task into regression.

5. The WasteSort problem

WasteSort accepts a photo containing one main waste item and predicts one of six materials: cardboard, glass, metal, paper, plastic or trash. We use TrashNet, a public dataset created by Gary Thung and Mindy Yang. Its original collection contains 2,527 labeled images. The pictures shown here are dataset examples. Similar-looking materials, such as glass and clear plastic, can make recognition difficult.

6. Supervised image classification in WasteSort

During training, the system receives photographs together with their material labels. The model learns visual patterns that help distinguish those labels. Later, during prediction, it receives a new photo without the answer and outputs a material category. We use separate images to check whether it learned useful patterns. The app uses a saved model to classify an upload. Uploading a photo does not automatically retrain that model.

7. What WasteSort can support

WasteSort demonstrates how machine learning can assist with material recognition. Its prediction can be wrong, especially with cluttered backgrounds, mixed materials or unfamiliar objects. A material label alone does not decide whether an item is recyclable or which campus bin is correct. Those instructions depend on verified campus rules. The main idea is that labeled images allow a supervised model to learn a material-classification task. Thank you.

Quick Review and Likely Questions

What is AI?

The broad field of systems that perform tasks such as recognizing images, reasoning and making decisions.

How is machine learning related to AI?

Machine learning is an approach within AI that learns patterns from data. Deep learning is a type of machine learning that uses multilayer neural networks.

Why is WasteSort supervised learning?

Each training image has a known material label. The model learns the relationship between the image and that label.

Why is it classification?

The output is one of six categories. Regression instead predicts a numerical quantity, such as travel time.

What is a label?

The known target answer for a training example, such as plastic or cardboard.

What is a model?

The learned mathematical system used to produce a prediction from an input.

What is the difference between training and prediction?

Training uses examples to adjust the model. Prediction uses the saved model on a new input. Uploading a photo does not automatically retrain it.

Why do we need separate evaluation images?

They help check whether the model can handle examples it did not learn from, rather than only remembering training examples.

What data does WasteSort use?

TrashNet by Gary Thung and Mindy Yang. Its original release contains 2,527 labeled photos in six classes. Cleaning may reduce the usable count.

Can it decide the correct campus bin?

It predicts material appearance. Bin instructions require verified campus rules and may also depend on contamination or mixed materials.

Are the complete research results part of this talk?

This talk covers the Monday concepts task. The full model comparison and validation belong to later research activities.

Sources

Google for Developers, What is Machine Learning? and Supervised Learning. ML introduction and supervised learning.

IBM, What Is Artificial Intelligence?. Thung and Yang, TrashNet. The group-supplied WasteSort proposal defines the project scope.

AI assistance: ChatGPT/Codex helped prepare this material. The presenter should verify and understand it before presenting.